

AI for Causal inference
Causal inference for AI

CAUSAL AI FOR HEALTH · KDDM WORKSHOP · SESSION W18

Vincent Jeanselme

Director of Machine Learning Research, Pathomiq
Affiliated, Columbia University

jeanselme.github.io

Disclosure of relevant financial relationships

The following presenter **has** relevant financial relationship with ineligible companies to disclose.

- Vincent Jeanselme – Pathomiq Inc., Salary and stocks

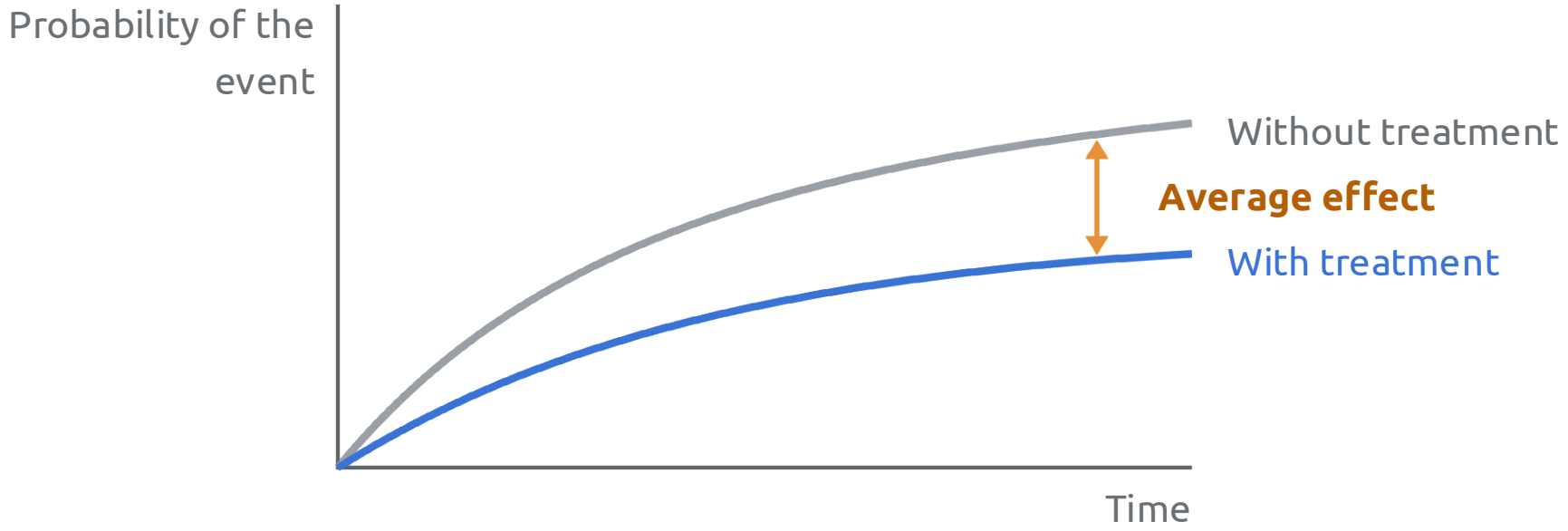
*All of the relevant financial relationships listed for these individuals have been mitigated.

AI for Causal inference

Causal inference for AI

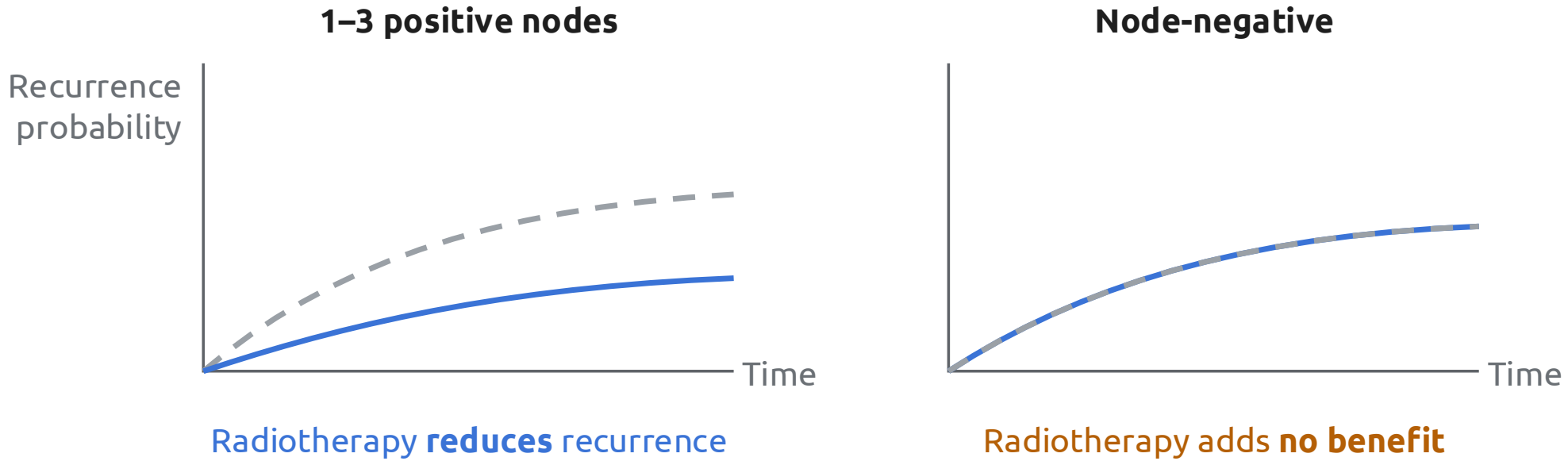
Jeanselme et al. (2025)
Machine Learning for Health (ML4H)
Identifying treatment response subgroups in observational time-to-event data

Average treatment effect



Randomized trials estimate whether a treatment works **on average**

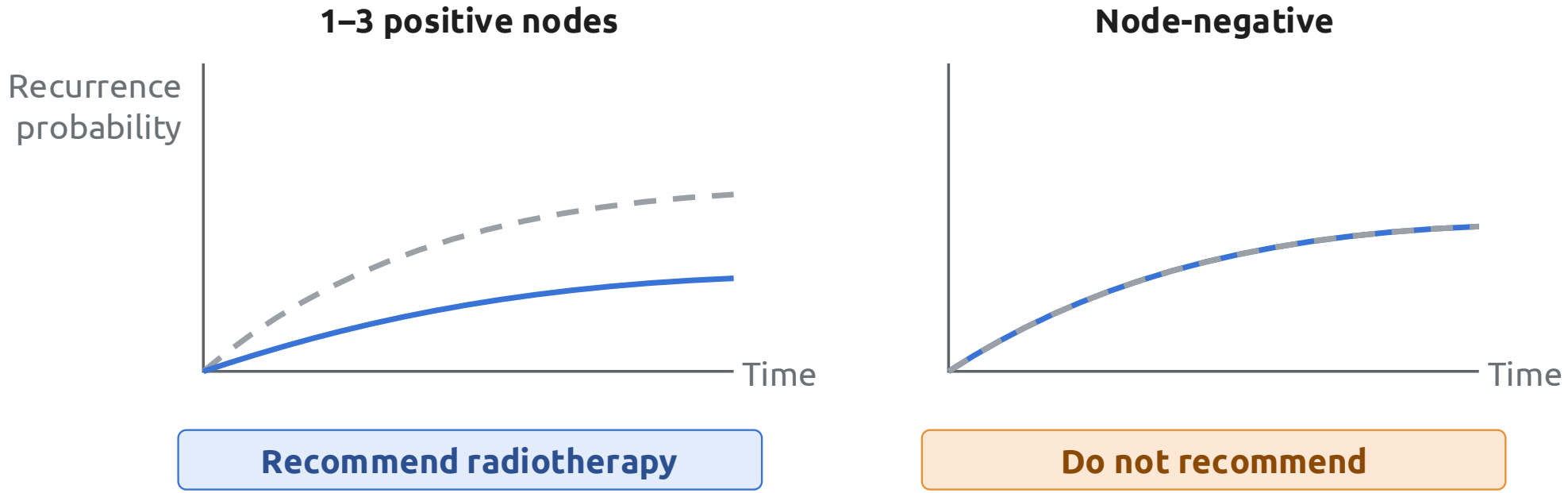
The average can hide different responses



Heterogeneous treatment effects: one treatment, different effects across subgroups

EBCTCG (2014), Lancet

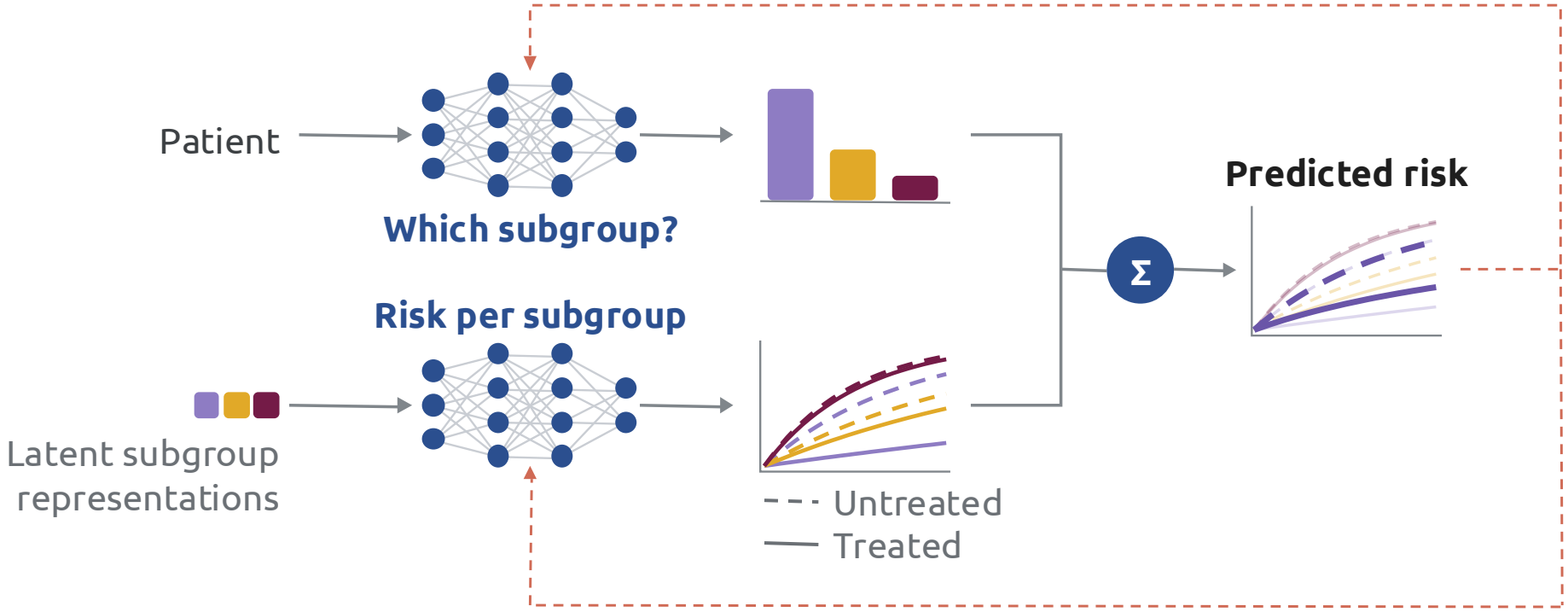
From heterogeneous effects to guidelines



How to identify heterogeneous treatment effects?

EBCTCG (2014), Lancet · ASCO/ASTRO/SSO guideline (2025)

Uncovering heterogeneous treatment effects with AI

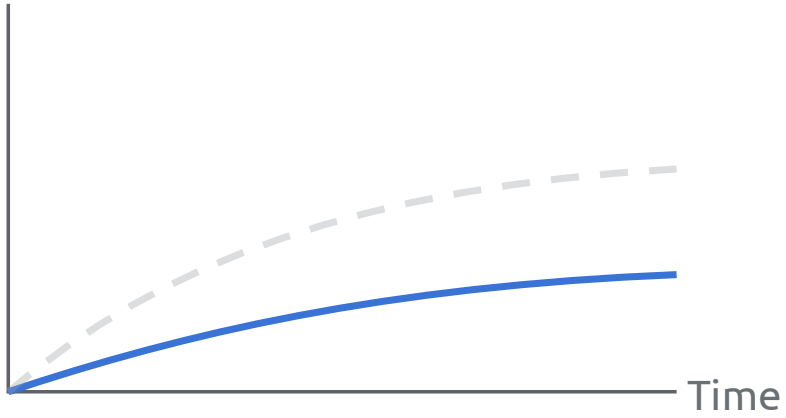


Trained **end to end**, so subgroups are discovered from observed responses

Jeanselme et al. (2025) *Identifying treatment response subgroups in observational time-to-event data*

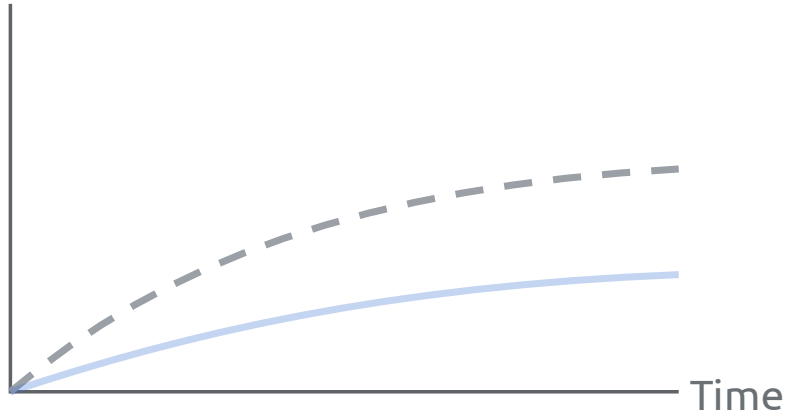
The challenge: one patient, one outcome

Treated patients



Observed **with** treatment

Untreated patients



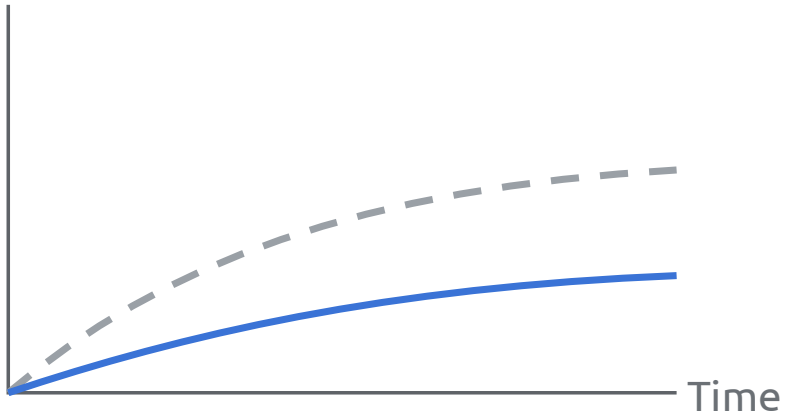
Observed **without** treatment

With treatment Without treatment

In simulation, our method **recovers** subgroups and their effects better than existing methods

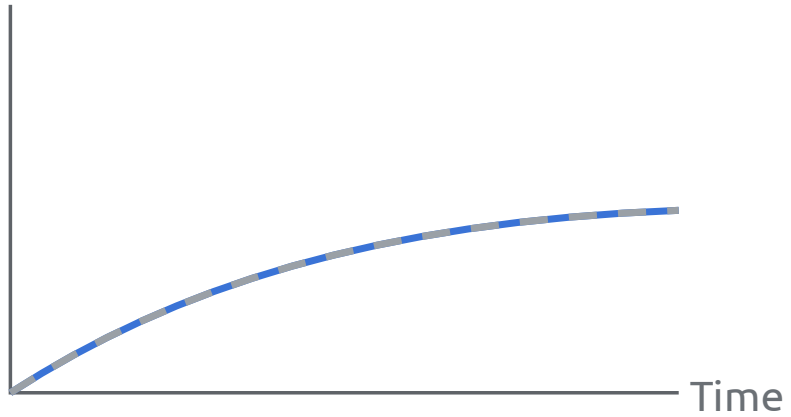
SEER: adjuvant radiotherapy in breast cancer

Many nodes, more HER2+ · 5.4%



Radiotherapy adds **+0.84 years**
over 5 years after diagnosis

Few nodes · 94.6%



Radiotherapy adds **+0.01 years**

239,855 women after chemotherapy
Results are hypothesis-generating

Jeanselme et al. (2025) *Identifying treatment response subgroups in observational time-to-event data*

Why AI for heterogeneous effects

- No need to pre-specify subgroups
- No proportional hazards assumption
- Flexible, nonlinear covariate effects
- Adaptable to any modality: imaging, notes, genomics

Jeanselme et al. (2025) *Identifying treatment response subgroups in observational time-to-event data*

AI for Causal inference

Causal inference for AI

Jeanselme et al. (2026)
JAMA Network Open
AI-generated draft response tool, message turnaround time, and after-hours replies among health care professionals

Clinicians' inbox today



Patient sends a message

Clinician writes the reply

Patient receives the reply

85 min Daily inbox time for primary care physicians

Arndt et al. (2017) Annals of Family Medicine

Deploying AI is an intervention



Patient sends a message

Language model drafts a reply

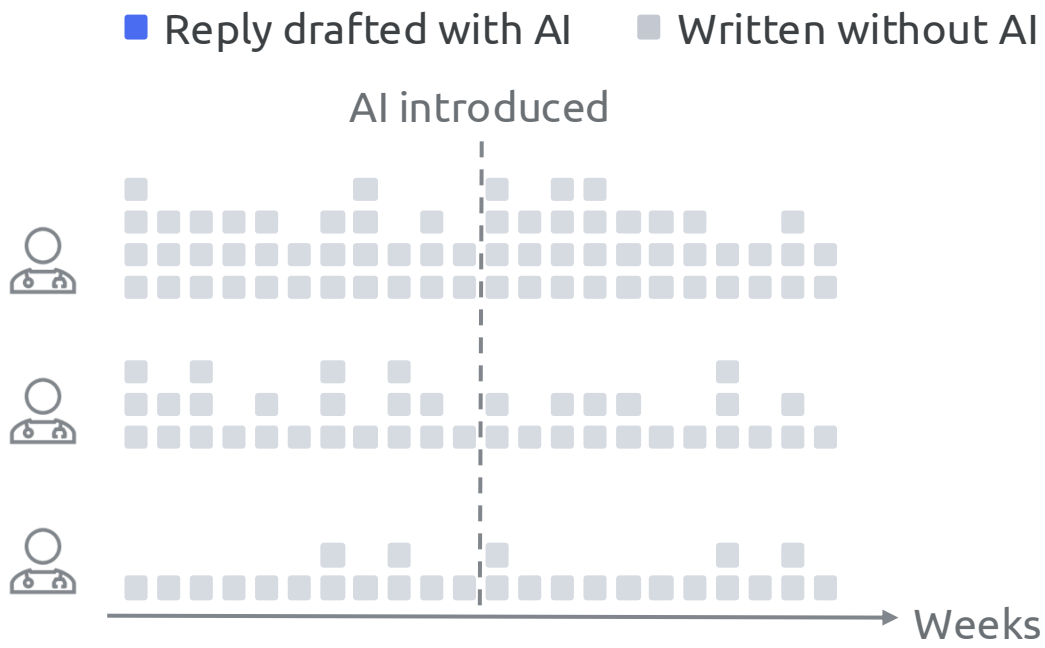
Clinician edits the draft or starts from scratch

Patient receives the reply

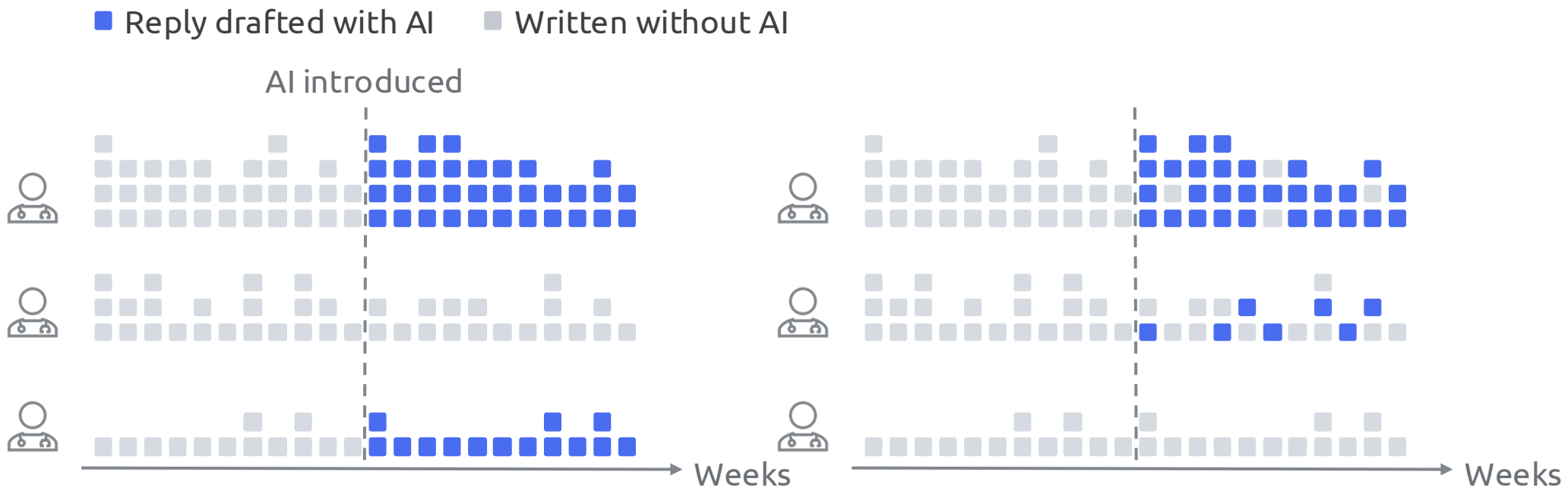
150+ Health systems have adopted the tool

Intervention	Introduce an AI tool into the hospital's workflow
Question	Does it impact clinicians' turnaround?

Bias 1 · Who uses the tool



Bias 1 · Who uses the tool

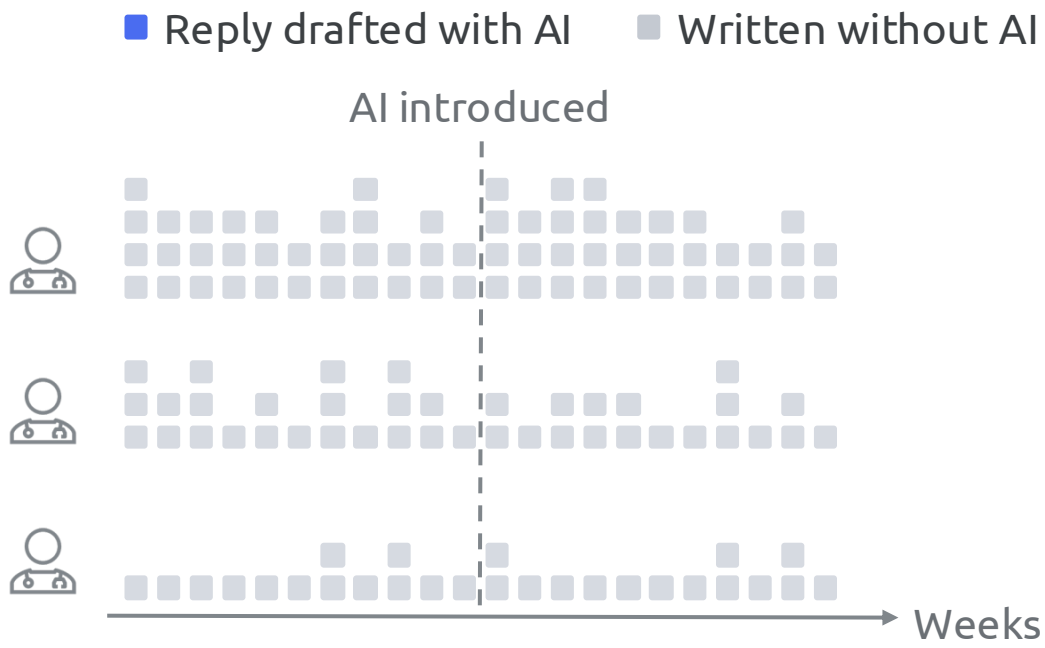


Randomized: AI given at random

In practice: busiest clinician may use AI more

AI effect | Clinician

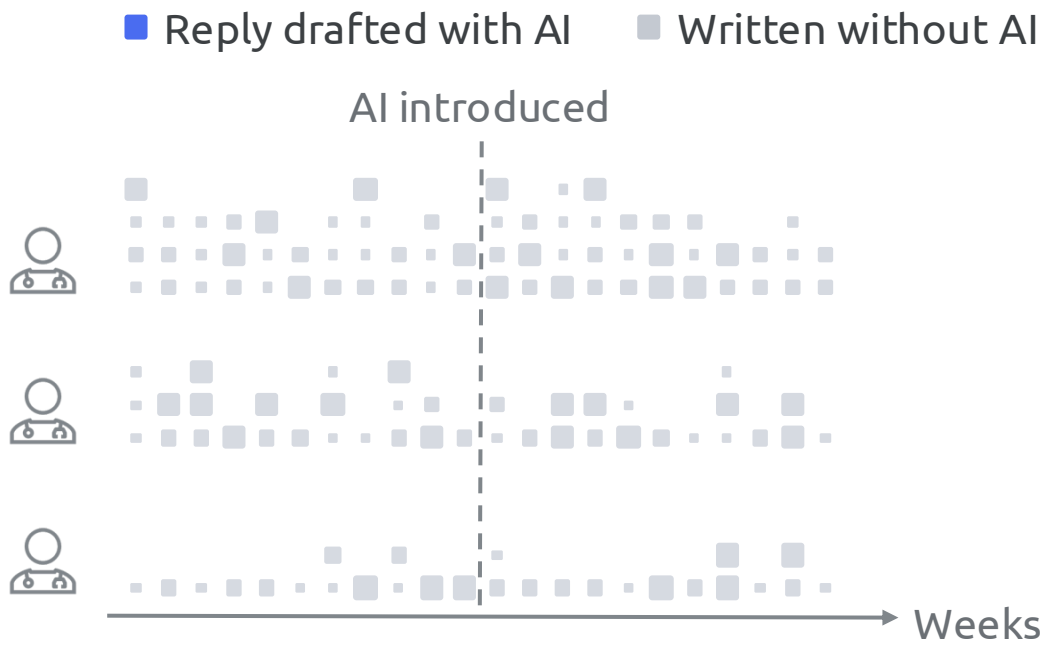
Bias 2 · Which messages



AI effect

Clinician

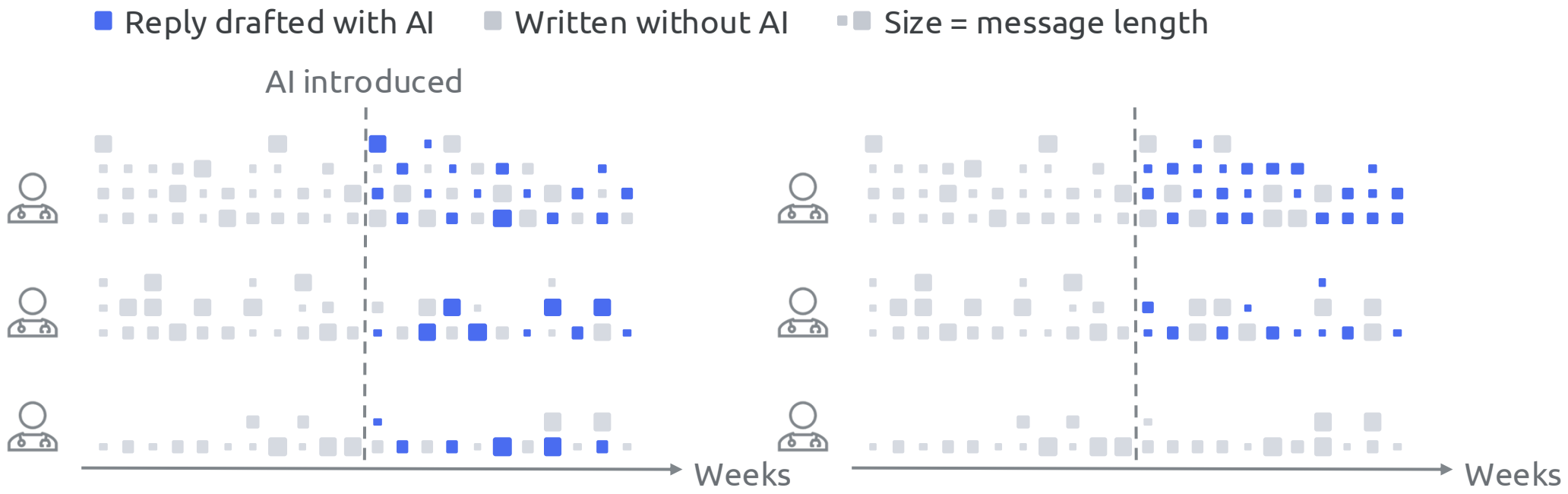
Bias 2 · Which messages



AI effect

Clinician

Bias 2 · Which messages

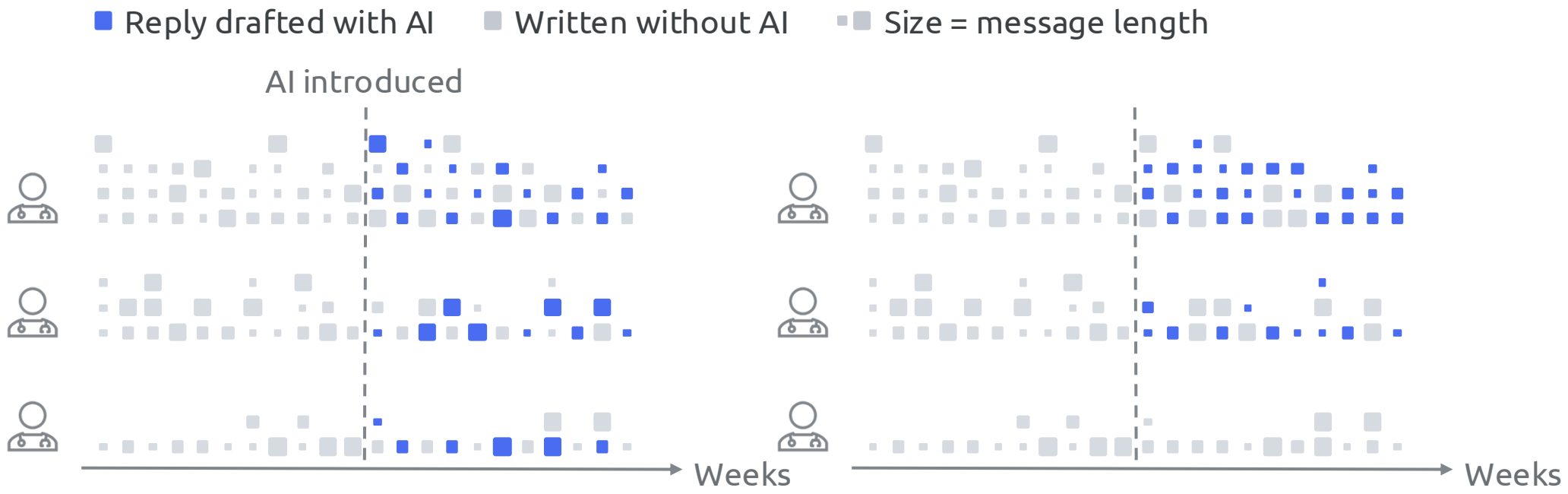


Randomized: AI used equally on all messages

In practice: AI may be used for shorter messages

AI effect | Clinician

Bias 2 · Which messages

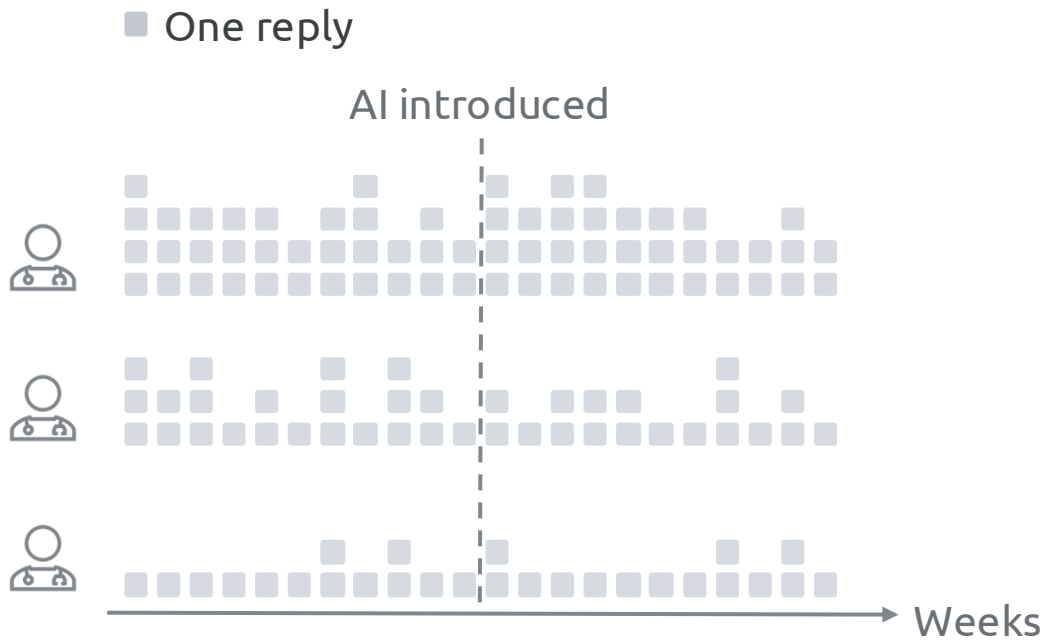


Randomized: AI used equally on all messages

In practice: AI may be used for shorter messages

AI effect | Message | Clinician

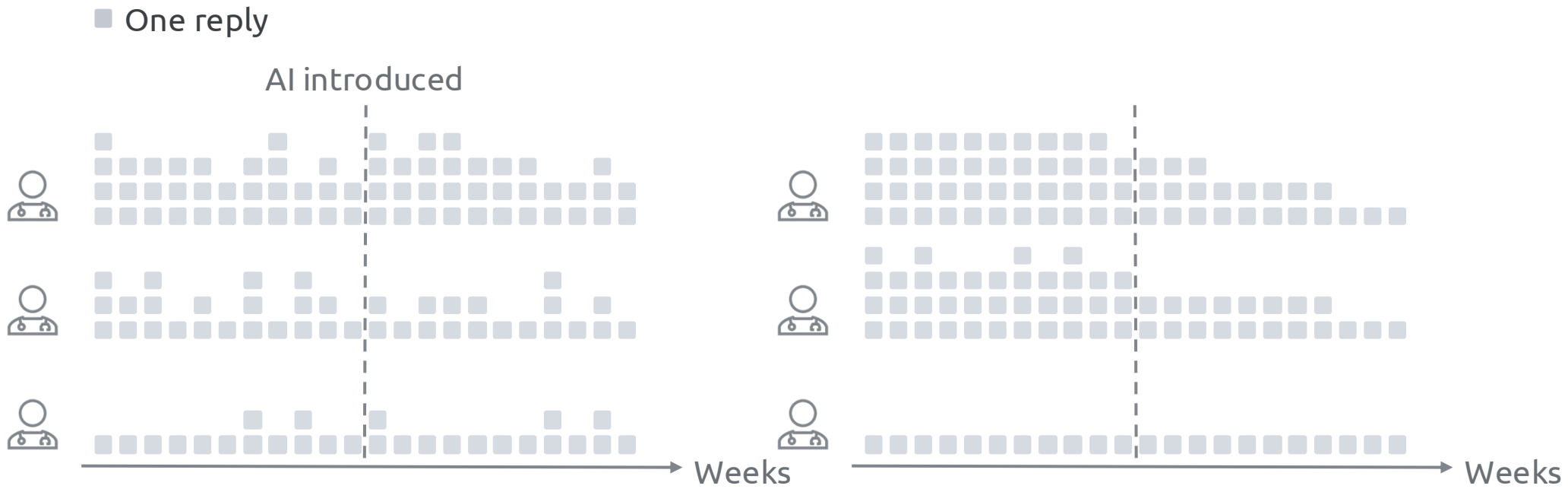
Bias 3 · Time



Ideally: comparable before and after

AI effect | Message | Clinician

Bias 3 · Time



Ideally: comparable before and after

In practice: trend impacting response time

AI effect | Time | Message | Clinician

Existing evidence

Study (n = users)	Design	Remaining biases		Main finding
UCSD · n = 52	Randomized rollout	Message		Longer reads, same reply time
Stanford · n = 162	Before vs after	Message	Time	No change, less exhaustion
NYU Langone · n = 75	Draft vs no draft	Message	Clinician	Turnaround 6.8% faster
CHOP · n = 323	Draft vs no draft	Message	Clinician	Response 13s faster
UMCG · n = 100	Draft vs no draft	Message	Clinician	No change

Proposed modeling

log(turnaround) = clinician fixed effect + calendar time effects
+ AI adoption + AI use + AI habituation + AI use × message characteristics

983,154 responses · **1,542** clinicians · **272,091** patients · Columbia University Medical Center

Jeanselme, Austin, Han, Lewis, Hruby, Natarajan, Joshi (2026) *AI-generated draft response tool, message turnaround time, and after-hours replies among health care professionals*, JAMA Network Open

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Clinician
Compare each clinician before and after adoption

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Clinician
Compare each clinician before and after adoption

Time
Absorb temporal trends

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Clinician
Compare each clinician before and after adoption

Time
Absorb temporal trends

Message
Differentiate the effect by message

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Findings

Turnaround time

+8.6% Adoption
95% CI 4.6 to 12.9

–29.5% Use: starting from a draft
95% CI –34.5 to –24.1

After-hours replies

~ 0 Overall
No overall reduction

–3.7 pp Results queries, working hours
95% CI –6.9 to –0.4

+4.5 pp Messages received after hours
95% CI 1.5 to 7.5

Introducing the tool appears to add friction,
but using a draft may **lower the barrier to responding**

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Why causal inference for AI evaluation

- Deploying AI is an intervention
- Who uses it, and when, is not random
- Design studies to evaluate its effect

Thank you

Questions?

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ML4H

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